

Generative AI and management education: a knowledge management perspective

Himanshu Rai, Nishtha Malik and Arup Varma

Abstract

Purpose – This study sought to add value to literature on the adoption of Generative Artificial Intelligence (GenAI) in management education through a knowledge management perspective. The study critically examines GenAI's dual potential to enhance and disrupt educational ecosystems, simultaneously serving as a catalyst for pedagogical innovation and a source of ethical and operational dilemmas. Drawing on complexity theory and paradox theory, the study aims to explore how academic institutions can manage the conflict between GenAI's transformative potential and the need to uphold honesty, integrity and transparency in education.

Design/methodology/approach – Semi-structured interviews were conducted with 67 stakeholders (directors, deans and students) at leading management institutes across India. Interview topics included (i) the pros and cons of GenAI usage in higher education, (ii) training students to effectively and efficiently use GenAI to solve real-world business problems and (iii) honesty, integrity and transparency in the use of GenAI.

Findings – The results reveal that integrating GenAI into management education requires (i) long-term strategic perspective, (ii) clear and adaptive policy frameworks and (iii) robust training mechanisms for all stakeholders. These steps would considerably mitigate the risks associated with GenAI adoption and help harness its potential in knowledge creation, analysis, dissemination and decision-making. Furthermore, the study reveals that the integration of GenAI in management education is characterized by paradoxes that warrant intelligent policy responses. Key recommendations include: (i) structured capacity-building for faculty and students, (ii) inclusion of GenAI in curricula with ethical regulations and (iii) the development of governance frameworks to foster responsible AI literacy across academic and industry boundaries.

Originality/value – The study examines the GenAI discourse through the dual lenses of Complexity Theory and Paradox Theory, an approach largely missing in current literature. While most studies have focused on the technological or ethical dimensions of AI in education, this study uniquely examines how business schools, as knowledge institutions, can strategically balance the benefits and risks of GenAI adoption. This study thus brings to light localized complexities, ethical ambiguities and actionable insights relevant to educators, practitioners and policymakers.

Keywords Generative AI, Management education, Large language models, Knowledge management

Paper type Research paper

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1. Introduction

The rapid evolution, wide application and quick acceptance of Generative Artificial Intelligence (GenAI) tools such as ChatGPT (Chat Generative Pre-trained Transformer) have led to major disruptions across sectors, including academics (see, e.g. Del Giudice, 2016; Del Giudice *et al.*, 2023). ChatGPT, an open source chatbot developed by OpenAI, possesses the ability to generate text in response to human prompt. It may be understood as conversational AI developed on a transformer-based architecture that has been pretrained on large data sets to generate human-like text. Large language models (LLMs)

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form the foundation of AI systems like ChatGPT, and models are trained on significantly large amounts of data to learn language patterns. The disruption that the introduction of GenAI caused across domains is reflected by the fact that ChatGPT acquired more than 100 million users within a span of two months after its launch (Peres *et al.*, 2023). Representing a broad group of AI systems, GenAI systems are complex in technical design as well as in practical implications. These platforms can generate new content, including text, images, music, code and video. Examples include Magic Write (Canva Docs) – produces text in response to human prompts; Eleven Labs (a deep learning model) – capable of transforming text into high quality audio in any voice and style; DALL-E 2, Midjourney and Stable Diffusion – capable of generating art, graphics, etc. on the basis of natural language descriptions.

The advent of AI and similar tools has led to extensive deliberation and debate on their appropriate use, particularly in the field of education. On the one hand, AI tools may help instructors streamline administrative tasks, prepare better lectures and deliver personalized learning; on the other hand, their use could raise questions about teachers' own commitment and ability, instructional freedom, data privacy and reliability of AI-generated content. Similarly, students may find AI tools very useful for instant access to information and benefit from AI's assistance with assignments, but this could also increase students' dependence on such systems, reduce critical thinking, inspire laziness and encourage academic dishonesty. Problems get compounded if irresponsible use of AI goes undetected in academics. For example, if output by GenAI goes undetected by plagiarism software (Gao *et al.*, 2023), the originality of submissions and the integrity of academic learning become questionable, defeating the whole academic endeavor. The key element in this context is responsible and honest use of AI (van Dis *et al.*, 2023).

While AI tool usage today seems unavoidable, it is also desirable as jobs in any domain in this day and age require knowledge of AI and other modern technological interventions. For example, students must, to begin with, know how to craft appropriate prompts (prompt engineering), learn to critically evaluate outputs generated by GenAI systems and understand the limitations, biases and reliability of such metrics and tools. Speaking of prominent professional domains that seek proficiency in AI tools, marketers may be required to use GenAI for search engine optimization, generating appropriate content for a website's landing page, social media copy or online reviews (Carlson *et al.*, 2023). Schweidel *et al.* (2023) have discussed ways in which GenAI can support marketing operations. Furthermore, image generators powered by GenAI can generate visually appealing content from textual prompts, enabling increased audience engagement and storytelling.

In finance, AI tools are used for automated financial modeling, risk analysis, fraud detection and market trend forecasting. GenAI can help in preparing preliminary analyst reports, summarizing earnings call transcripts, or generating investment memos using real-time financial data. In human resource management, AI tools may be used in automated resume screening, candidate engagement through chatbots, employee sentiment analysis and training content creation through GenAI tools. AI tools help with demand forecasting, route optimization, inventory tracking and supplier risk assessment in operations and supply chain management. Predictive AI models are used to streamline logistical decision-making and minimize disruptions. GenAI is also used in domains such as entrepreneurship, business strategy, analytics and decision science. It is clear that AI tools are becoming key enablers of professional capability across domains such as marketing, finance, operations, HR and strategic management, demanding responsible and informed usage by the next generation of business leaders.

It is noteworthy that despite all advancements in AI systems, human oversight and intervention remain critical elements. AI-generated output may carry factual and logical inaccuracies, the identification and removal of which require human involvement.

Reisenbichler *et al.* (2022) noted that there remains the need for human monitoring for rectifying any inaccuracies or brand related errors in the context of AI-generated output. While the authors have discussed above the issues that surround AI use, and also some of the domains that presently require students to have knowledge of AI tools and their applications, educational establishments should lay down clear rules and policies in relation to AI use and train students to employ them effectively and ethically (Gupta *et al.*, 2024; Mollick and Mollick, 2023; Rana *et al.*, 2024).

This study adopts a dual-theoretical framework drawing from Complexity Theory and Paradox Theory to better understand the multifaceted implications of integrating GenAI into management education. Complexity Theory allows for an examination of educational institutions as dynamic and interconnected systems, the responses of which to disruptive technologies like GenAI are nonlinear, emergent and dependent on context (Donaldson *et al.*, 2025). Paradox Theory, on the other hand, emphasizes the lasting conflicts or tensions among elements of institutions when efforts are made to balance seemingly contradictory but interdependent goals, for example promoting innovation while ensuring academic integrity, or enabling personalized learning without fostering dependence (Lim *et al.*, 2023). These theories provide an appropriate and nuanced lens to analyze how educators and institutions must face and manage the unpredictable and often conflicting outcomes associated with the adoption of GenAI.

While interest in GenAI adoption in the education domain is increasing, significant gaps in extant literature remain. This study helps address three such key gaps:

1. Lack of strong theoretical frameworks, as several existing studies focus mainly on the technical or surface-level aspects of GenAI; the use of deeper academic theories to understand its impact on sectors like education remains ignored. To the best of the authors' knowledge, no study has drawn from Complexity Theory or Paradox Theory to understand the issues associated with the adoption of GenAI in educational environments. This study fills that gap by using both these theories to offer a better understanding of how institutions are and should respond to disruptive developments like GenAI.
2. Missing perspectives from non-western education systems. As most research on GenAI in education comes from Western countries. There is very little detailed research showing how GenAI is being contemplated upon, used or challenged in countries like India. This is a major gap, considering the fact that India has one of the largest and most varied and vibrant management education systems.
3. No clear guidelines for ethical GenAI use in practice, it is widely acknowledged that GenAI in education should be used ethically.

However, it remains unclear how schools and colleges can put this into practice. The present begins to bridge that gap by connecting GenAI use with responsible governance and meaningful value-based education practices. In light of the discussion above and considering the seemingly opposing sets of outcomes that GenAI use carries, especially in relation to integrity, honesty, effectiveness, assessment processes, this study addresses the following research questions mentioned below by conducting in-depth interviews with different stakeholders involved in the process:

- RQ1. How to manage a clear discourse with students about GenAI use with integrity, transparency and honesty?
- RQ2. How to train learners to effectually use these tools for resolving real-life business problems, including training on "prompt engineering"?
- RQ3. What assessment designs or types can be most effective in integrating GenAI into the assessment process?

RQ4. What is the impact of GenAI on academic integrity policies?

In light of these questions, the study reveals that integrating GenAI into management education demands clear and continuous communication between faculty-members and students to ensure integrity, honesty and transparency in usage (Erdem Coşgun, 2025; Gonsalves, 2024). Structured training on prompt engineering, experiential projects and academia–industry collaborations emerged as a valuable medium to equip learners with real-world problem-solving skills using GenAI tools. Furthermore, the need for constant evolution of assessment design to encourage critical thinking and creativity is highlighted (Budhwar *et al.*, 2023; Luckin *et al.*, 2016; Usher, 2025). The study also emphasizes balance between innovation and ethical governance through comprehensive policies that integrate AI literacy, disclosure norms and faculty training (Eke, 2023; Rana *et al.*, 2024). Through these insights, the study demonstrates that the integration of GenAI in management education indicates, beyond just a technological disruption, a systemic and ethical change in the educational domain that educational institutions must strategically manage (Gupta *et al.*, 2024).

The rest of the paper is structured as follows: The next section offers a theoretical background of the study, explaining how the study draws from Complexity Theory and Paradox Theory to examine the integration of GenAI into management education. It is followed by literature review and methodology adopted by the study. Next, findings and discussion sections are laid down. After the discussion section, implications for teachers and students are presented, followed by policy and institutional recommendations. Limitations and future research scope, and conclusion form the final sections of the study.

2. Theoretical background

Considering the conflicting realities of GenAI use (particularly in the management education context) and inherent complexities therein, the present study draws from Complexity Theory and Paradox Theory to better understand the processes and consequences associated with integrating sophisticated systems like GenAI into management education. Complexity Theory sees organizations as complex systems comprising interconnected, codependent and evolving elements the responses of which to internal and external stimuli are not necessarily linear (Anderson, 1999).

This theoretical approach offers an appropriate method to observe the unpredictability and emergent nature of change when disruptive technologies like GenAI are introduced into educational systems (Donaldson *et al.*, 2025; Kolt *et al.*, 2025; Susnjak *et al.*, 2025).

Paradox Theory, on the other hand, emphasizes that in a given system, apparently interrelated but contradictory elements exist and persist over a period of time (Yap and Lim, 2023). Examples include stability and change, innovation and regulation, efficiency and creativity, etc. In the context of this study, some examples could be innovation and integrity, efficiency and critical thinking, guidance and dependence, among others. According to this approach, these dichotomies cannot be resolved, but managed (Cunha and Putnam, 2019). Paradox Theory thus aptly addresses the lasting contradictions that emerge in educational systems when such establishments attempt to balance conventional pedagogical values with transformative technological developments like GenAI (Lim *et al.*, 2023; Schaap and Vanlommel, 2024).

These theories thus provide a sound grounding for the current study pertaining to the adoption of GenAI in management education. An educational institution is indeed a complex system with seemingly contradictory but interlinked components. The introduction of a disruptive factor like GenAI may beget outcomes that cannot be accurately predicted or fully controlled. Furthermore, the outcomes may not necessarily be linear in relation to their causal factors (extent of use, institutional readiness, level of control, pedagogical intent, etc.).

The following illustration validates the application of Paradox Theory and Complexity Theory in this study: some educators may welcome and adopt GenAI due to their view that such technological advancements would enhance instructional design and student engagement and assist students that need personalized learning techniques for better retention. However, other instructors may resist due to concerns about academic integrity or loss of humanistic learning values. Here, Complexity Theory enables us to study these opposing reactions as lasting elements that form part of a complex system and its processes influenced by localized contexts. At the same time, Paradox Theory helps us understand the inherent conflict educators face when they are confronted by the need to simultaneously ensure several outcomes: promote critical thinking while employing AI's efficiency; uphold assessment integrity while opening up to personalized learning; encourage innovation while enforcing control; inculcate learning integrity and work ethic in the presence of readily accessible and easier alternatives.

It is noteworthy here that these conflicts or tensions cannot be completely resolved but are to be managed. Acknowledging and accepting these paradoxes within complex systems would enable educational leaders to better understand and more effectively face the challenges posed by the introduction of disruptive advancements like GenAI in educational institutions. Thus, the dual-theory approach adopted in this study offers an effective way to examine the domain of GenAI-enabled education.

3. Literature review

Extant literature, on the one hand, acknowledges the likely advantages of AI in boosting learning, personalizing instruction and streamlining educational administration; on the other hand, it remains inconclusive on its impact on pedagogical processes and assessment integrity.

AI has evolved into “computing systems that are able to engage in human-like processes such as learning, adapting, synthesizing, self-correction and use of data for complex processing tasks” (Popenici and Kerr, 2017, p. 2). Rapidly evolving LLMs are becoming increasingly capable of responding more intelligently than their predecessors based on dialogue-based or rational exchanges with users (Ouyang *et al.*, 2022). AI today is being employed to assess student engagement and academic integrity (Gonsalves, 2024); it can produce tailored assessment activities on the basis of specific individual student needs (Zawacki-Richter *et al.*, 2019). It also offers process-oriented assessment of collaborative performance (Kim *et al.*, 2022).

AI use may offer several benefits, including help in determining the most advantageous learning methods, timely intervention, evaluating and monitoring student improvement and rendering the teaching process more engaging and interactive (Celik *et al.*, 2022). Studies report several possible AI integrated learning designs oriented toward multidisciplinary learning, creativity and problem solving (Kim *et al.*, 2022), exploring various poetic qualities (Kangasharju *et al.*, 2022), or optimizing learning outcomes through proximal development (Ferguson *et al.*, 2022). According to Castañeda and Selwyn (2018), AI has the capability to revolutionize education, but the mere use of AI systems is not sufficient to achieve positive educational outcomes. Bower (2019) emphasized the use of technology in close association with learning and educational theories to direct teaching design and high-tech progression.

In the context of AI-supported learning, Zheng *et al.* (2021) underscored the use of AI in Education (AIEd) technologies in connection with learning content and instructional methods and emphasized the integration of these to augment learning efficiency and efficacy. Furthermore, six key elements must be considered to achieve expected learning objectives when devising AI-supported learning activities: learners, context, tasks, pedagogic methods, methods of interaction and AI technology application

(Zheng *et al.*, 2021). Studies mention several benefits of using AI in education for students, including higher levels of engagement and motivation (Xia *et al.*, 2022), increased interest in learning (Lin and Chang, 2020), facilitation of environments of interactive learning (Karsenti, 2019), improvements in academic performance (Khan *et al.*, 2021; Kim *et al.*, 2021), enhanced learning results and student happiness (Winkler and Söllner, 2018), maximized student learning capabilities (Clarizia *et al.*, 2018), tailored learning experiences (Cunningham-Nelson *et al.*, 2019), critical thinking and creativity and evaluation of intricate skills (Luckin *et al.*, 2016), deeper thinking (Chiu *et al.*, 2023), improved communication capabilities in the contexts of language learning (Vazquez-Cano *et al.*, 2021), better peer communication skills (Hill *et al.*, 2015) and others.

Benefits of AI use for teachers and administrators include positive attitudes (Aldosari, 2020), improved instructing capabilities (Jaiswal and Arun, 2021), enhanced teaching proficiency through inspiration and encouraging self-contemplation, adjustable instruction approaches (Aldeman *et al.*, 2021), enhanced teacher insight into students' learning processes (Vincent-Lancrin and Van der Vlies, 2020), professional development (Gunawan *et al.*, 2021; Hu, 2021), increased ability to analyze a learner's learning capability (Durall and Kapros, 2020) and others. For administrators, AI can help address disengagement and reduce dropout rates among students (Karsenti, 2019; Villegas-Ch *et al.*, 2021), predict and identify risks, replace and reduce repetitive work, (Chan and Zary, 2019), provide feedback to learners (Chen *et al.*, 2020; Cunningham-Nelson *et al.*, 2019), offer customized learning, assist students, expedite completion of administrative tasks and help in assessment (Hopcan *et al.*, 2022).

However, despite the acknowledged possible advantages of using AI in education, studies have reported major concerns, such as strong indications that assessments could be compromised relatively easily using GenAI, rendering it difficult to determine which were human-authored, GenAI-modified or GenAI-authored. Furthermore, AI might just improve access to content and do less to actually change teaching and learning methods (Zhai *et al.*, 2021). A key element to be considered is the way students engage with AI [as passive recipients (AI-directed), active collaborators (AI-supported) or empowered leaders of their own learning (AI-empowered)] (Ouyang and Jiao, 2021). Swiecki *et al.* (2022) listed several challenges that required addressing when employing AI in assessment tasks, including ignoring expertise, lack of accountability, increased surveillance-based teaching and a damaging human-machine divide in the assessment process.

Hisan and Amri (2023) cautioned against problems related to identity, plagiarism and assurance of learning arising due to increased access to GenAI tools. The purpose of education itself is defeated if students become able to pass evaluations by submitting work that they did not themselves accomplish (Cotton *et al.*, 2024). While various AI detection tools have also proliferated and are being increasingly employed, such systems may be described as more probabilistic and not accurate, and these tools may not be able to help evaluators truly determine whether an assignment submitted by a student is done by the students themselves (Cotton *et al.*, 2024). Rapp *et al.* (2021) cautioned against assessing a chatbot merely on the basis of its efficacy, utility and capability to fulfill and engage individuals. The practice gave rise to ethical issues associated with possible misuse and deception as users could erroneously trust that they were interacting with actual people rather than with Chatbots. Apprehensions about how ChatGPT "might manipulate and deceive people" have been recorded (McCallum, 2024).

As GenAI tools become more accessible and are being increasingly used by students, Chan (2023) argued that much needs to be done to lay down comprehensive and focused policies on AI in the domain of education. Initially, the use of GenAI was proscribed for students in certain universities, but this approach came under criticism as it was seen as unsustainable and backfiring in developing future-ready AI-literate citizens (Sullivan *et al.*, 2023). Chan (2023) proposed a GenAI policy framework highlighting pedagogic,

operational and ethical dimensions to be considered by higher education stakeholders when formulating policies.

Organizations such as the United Nations Educational, Scientific and Cultural Organization (UNESCO, 2023) and the Organization for Economic Co-operation and Development (OECD, 2023) have laid down guidelines for regulation of GenAI in education. However, these procedures ignore certain difficulties faced by higher education evaluation. Research on AIEd often doesn't focus enough on teaching methods and learning strategies. It also tends to ignore the many challenges and risks that come with using AI in classrooms (Guilherme, 2019). Goel and Polepeddi (2018) emphasized equity and accessibility for students. AI-based educational tools must ensure privacy of users and security of data and not support biases or spread disparities. Efforts have been made to regulate the use of AI in the education domain. For example, The University of Leeds employs a three-tiered classification (traffic light system) for GenAI use, where red means "no AI use permitted," amber signifies "limited, assistive AI use allowed," and green permits "fully integrated AI use encouraged" (University of Leeds n.d.). This structured traffic light approach is commonly in use internationally to uphold academic integrity, remove ambiguity around acceptable AI use and promote transparent dialogue between educators and students (University of Leeds n.d.; University College Dublin, College of Arts and Humanities n.d.), and to ensure that for each piece of assessment, staff and students have a shared understanding of whether Gen AI can be used and if permitted how, how much and where in the assessment process (Lancaster University n.d.). The three-way categorizations are devised to clearly communicate to students what is and is not permissible regarding GenAI use.

While the debate on the ethical, efficient, effective and correct GenAI usage in teaching is inevitable and will continue, it is equally true that academic institutions must embrace these technologies as a crucial element of study and instruction (Eke, 2023). It can be logically and safely deduced that in times to come, the use and impact of GenAI tools in education will only grow. Therefore, continual research is needed to determine the benefits and appropriate use of these technologies in education (Chiu *et al.*, 2023; Hwang and Chang, 2021). In light of the literature review above, it may be said that literature offers inconclusive evidence with respect to the impact of GenAI on assessments (Bower *et al.*, 2024). It can, however, be safely proposed that GenAI use does affect assessments and it becomes difficult for evaluators to determine the extent thereof. Though the effect of AI on education remains ambiguous (Holmes *et al.*, 2021), it shows promise to augment learning, academic innovations, instruction, assessment and educational administration (Chen *et al.*, 2020).

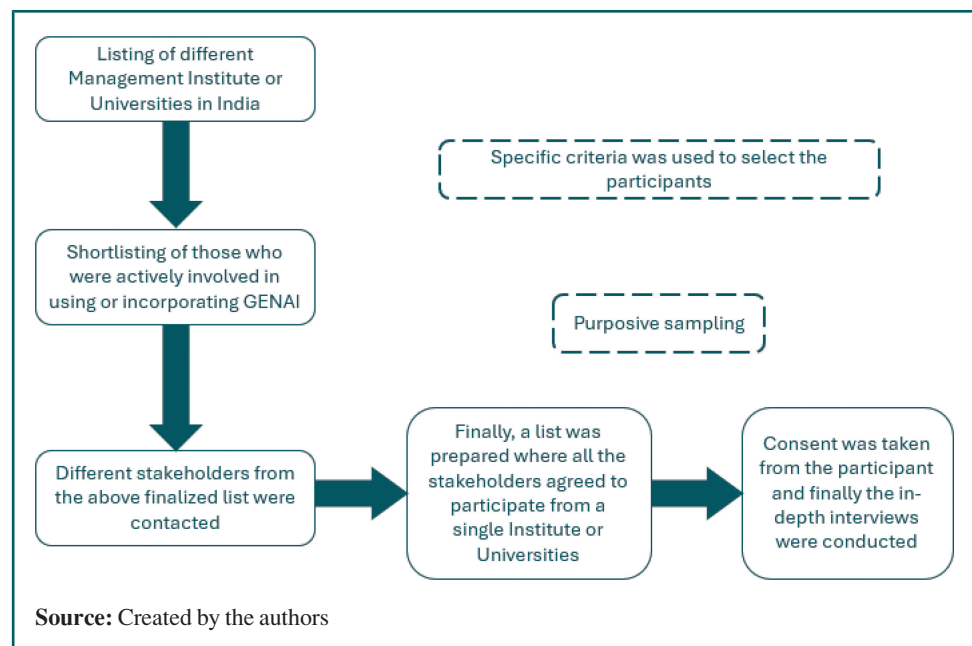
Furthermore, existing literature offers evidence that educational ecosystems are evolving in the presence of GenAI, and that the teaching and assessment domain is not linear or predictable, but complex, adaptive and surrounded by interdependent variables. This aligns with Complexity Theory which emphasizes the recognition of the nonlinear, emergent and context-sensitive nature of educational change. At the same time, the promise and risks associated with GenAI's application in education reflects the essence of Paradox Theory where institutions must face seemingly contradictory requirements: innovation and integrity, efficiency and depth and automation and human agency. It is here that the significance of the present study is highlighted. While studies document the functional uses and ethical dilemmas of GenAI, there remains a critical gap in examining how institutions may effectively respond to this paradox-laden complexity. Our study seeks to address this gap through the lenses of the two theories to assess how GenAI can be meaningfully integrated into teaching and assessment practices without compromising core pedagogical values. In doing so, we contribute significant value to the body of knowledge on GenAI in the education and assessment domain, proposing that GenAI is a complex phenomenon to be thoughtfully and carefully accepted, and effectively managed through continual institutional reflection, policy innovation and pedagogical reimagination.

4. Method

In this study, a qualitative approach was used where semi-structured personal interviews were carried out to collect data from directors, deans, teachers and students at different academic institutes across India (e.g. Gioia *et al.*, 2013). Qualitative methods accept the plurality of realities, support the formation of ideas and holistic insights and enable researchers to explore a given phenomenon in great detail. Hence, this approach was well-suited to effectively address our research context and questions (Denzin and Lincoln, 2011; Köhler, 2024; Alshwayat *et al.*, 2021). An inductive and interpretive approach was employed, given the exploratory nature of the subject and because this approach provided for the exploration of deeper patterns and rich insights into the lived experiences of the participants in relation to the phenomenon being studied (Corriveau, 2020; Piekkari *et al.*, 2022).

Initial respondents were identified using purposive sampling to ensure a diverse and representative group of stakeholders in the management education community. As highlighted by Patton (2002), purposeful sampling is a key strategy in qualitative research as it assists in the search for information-rich cases which can be studied in-depth (De Gagne and Walters, 2009). This sampling method remains unique to the purpose of each synthesis and useful in attaining data saturation and sufficiency (Patton, 2002; Suri, 2011). Selection criteria for participants were carefully laid down; only those institutes or universities were included where GenAI had been incorporated into processes (detailed selection process is given in Figure 1). To ensure appropriate representation, efforts were made to include participants who could assist in capturing diverse perspectives and experiences, resulting in robust and credible findings. Following Köhler (2016), data source triangulation was incorporated into the study's qualitative research design to achieve methodological rigor. Efforts were made to gather information from all stakeholders, including directors, deans, faculty-members, course instructors and students. This is in line with the works that discuss best practices for qualitative research in the field of education and knowledge management (Barbera *et al.*, 2015; Köhler, 2016; Sithipolvanichgul *et al.*, 2025). Out of the 210 respondents approached, 78 gave their consent to participate in the

Figure 1 Selection process



study, and finally, a total of 67 respondents were included, of which 23 were directors, deans and teachers while the remaining 44 were students from various disciplines. The participants were guaranteed confidentiality of data and anonymity (Lune and Berg, 2017). The findings and interpretations of this study are based on the responses from the top-ranking management institutes in India; therefore the differences in operational, cultural and regulatory contexts may limit the generalizability of the findings of this study.

Topics covered in the interviews included: major issues concerning GenAI usage in higher education; integration of GenAI in higher education and the potential risks associated with it; managing explicit discourse with students regarding the use of GenAI with unambiguity, veracity and honesty; training learners to effectually use these tools for resolving real-life business problems (training on “prompt engineering”); GenAI's impact on teaching and learning; etc.

To establish validity and reliability, we maintained:

- *credibility* by adopting established research methods, i.e. in-depth interviews;
- *confirmability* by continuing with the interviews until saturation was reached and similar themes were reported by the participants; and
- *dependability* by involving qualitative experts in the interview process and also in the reporting process.

This is in line with the recommendations of Guba (1981) and Patton (2002).

Based on the willingness and availability of the respondents to participate in the study, personal interviews were conducted which lasted from 40 min to just over an hour (average 50 min). All the interviews were transcribed and validated by the participants. Furthermore, the responses for separate research questions were organized in distinct files; the data therein was processed to remove the special characters, numeric values, spaces and so on. Uniformity of cases was accomplished for clean files by removing stop words. These files were then imported into the software (NVivo) for data analysis and converting the code into themes. Data was analyzed manually also, as suggested by Braun and Clarke (2006), the files were thoroughly read and re-read to generate aggregate dimensions from first-order and second-order descriptive concepts. All relevant responses were read line by line many times to generate different codes known as first-order descriptive concepts. This process is in line with the principle of contextualization, which emphasizes understanding the social and historical background of the research setting in a given qualitative study (Klein and Myers, 1999). The following issues were identified through descriptive coding: focusing on actual learning of objectives of teachings, generating learning materials with GenAI tools, designing quizzes for assessment purposes, augmenting learning with GenAI, exhorting students to use the GenAI tools, being aware of the law and regulations of the university or the institute, contemplating learning goals, using GenAI tools for writing and learning purposes, encouraging coding with the help of GenAI tools and identifying potential risks associated with GenAI tools. Overlaps were reduced by the creation of concepts or themes that were mutually exclusive.

Initial codes (first order) were rooted in narratives based on respondent-centric terms. Once both – emerging codes and respondent-centric terms – were identified, similarities between them were determined and noted (Gehman *et al.*, 2018; Smith and Beretta, 2021). Next, second-order concepts were created where themes were generated on the basis of extant literature on GenAI and descriptive concepts created in the first step (Gioia *et al.*, 2013). This step assists in generating second-order themes rooted in identified links by aggregating first-order codes. This data structure not only provides a meaningful visual aid but also demonstrates graphically the progression of the entire process, i.e. how the raw data provided by the respondents are converted into first order concepts, then into second

order and finally into aggregate dimensions. This is the most crucial step toward validating rigor in qualitative research as it enables researchers to draw theoretical insights for the study (see Figures 2 and 3). Through collaborative efforts of the research team across the process of determining codes and coding, Inter-coder reliability at 90% level of agreement was reported; group consensus-based approach was used to establish face validity.

5. Findings

Our study revealed that integrating Gen AI into management education would assist both students and academicians. Key findings are categorized below under several aggregate dimensions.

5.1 Dimension 1: Ways to conduct explicit discourse

Theme 1: Knowledge and awareness about the usage of GenAI

Theme 2: Preventive/corrective information and measures

Theme 3: Discourse on risk and benefits

Senior academicians believed in having an explicit discourse with students with respect to the use of GenAI with integrity, transparency and honesty. During such discourses, ethical guidelines should be laid down to the students, emphasizing responsible usage. Students should be asked to specify the usage of GenAI in assignments in terms of percentage. Most respondents also highlighted that preventive/corrective information and measures like plagiarism check applications could help students and academicians follow a clear structure in GenAI usage. Leading by example, repetitive discourse on risk and benefits to promote critical thinking, addressing ethical dilemmas and reinforcing the importance of responsible GenAI use were reported as other key elements.

The director of one of the management institutes noted [R03]:

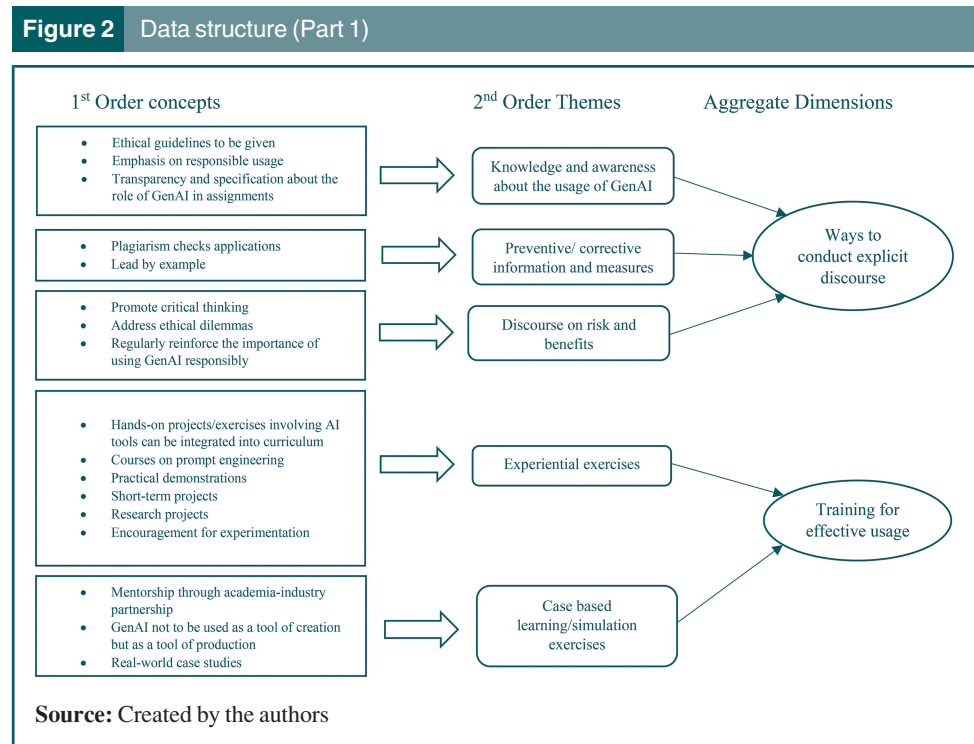
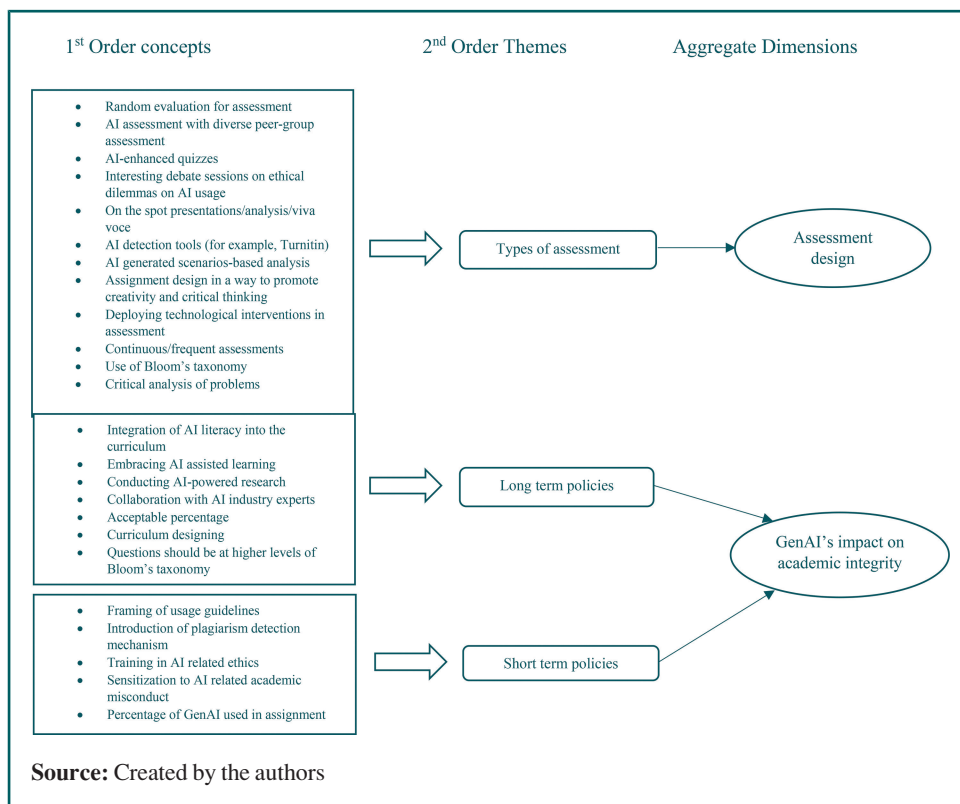


Figure 3 Data structure (Part 2)

Fostering a learning environment where students are well-informed and equipped to uphold academic integrity while harnessing the advantages of AI technology. Declaration from the student about using AI-enabled tools in the assignments.

A senior academician with 35 years of experience mentioned the following [R06]:

We need to discuss its purpose and ethical guidelines, emphasize responsible use, be transparent about its role in assignments, encourage collaboration and feedback, stay updated on AI developments, led by example, promote critical thinking, address ethical dilemmas, respect privacy, and regularly reinforce the importance of using Gen AI responsibly, fostering a learning environment where students are well-informed and equipped to uphold academic integrity while harnessing the advantages of AI technology.

5.2 Dimension 2: Training for effective usage

Theme 1: Experiential Exercise

Theme 2: Case based learning/simulation exercises

Most respondents named training on prompt engineering, assignments, short-term courses on prompt engineering as the means to educate learners to effectually use AI tools to solve real-life business problems. They highlighted the need to encourage students to refine and experiment with prompts to consistently achieve the desired results. Real-world case studies and hands-on projects involving AI tools should be integrated into the curriculum to allow students to gain practical experience with these tools for business problem-solving. The institute should start with a strong foundation in AI fundamentals, followed by practical application. Initially, the basics of AI algorithms,

machine learning and natural language processing must be taught. Then, the importance of understanding the problem domain, defining clear objectives and crafting well-structured prompts that guide the AI tool's output must be emphasized. Some respondents suggested having case-based learning/simulation exercises where mentorship through academia-industry partnerships could be included. Students should be explained that GenAI should be used responsibly.

Some responses are quoted as follows:

Courses on prompt engineering, practical demonstrations, short-term projects, research projects, Design Quizzes for Assessment, Augment Learning with GenAI, etc. are some ways that can help. [R04]

The case studies and research projects should be relevant to the needs of the respective industries. Mentorship of learners through academia-industry partnerships and interdisciplinary collaboration can create an atmosphere of holistic learning and generate innovative applications of Generative AI. Instilling the skill of creating precise prompts through courses on "prompt engineering" can help students derive more meaningful insights from AI utilization. [R08]

We can train students by explaining the fundamentals of AI and business problem-solving processes. We can train students by providing hands-on training on AI tools and platforms in the business setting. We must tell students how to collect and clean (pre-process) data for any AI business applications. We must explain prompt engineering, maybe using different public domain applications/tools. We have to discuss the role of algorithms in decision-making and industry-specific applications of AI in business. We may implement project-based learning to solve real-world problems. [R10]

5.3 Dimension 3: Assessment design

Theme 1: Types of assessment

Respondents emphasized that assessment designs promoting creativity and critical thinking could help effectively integrate GenAI into the assessment process. Random evaluation, continuous/frequent assessments and exciting debate sessions on issues involving ethical AI usage were some suggestions given. Assessment strategies must take into account the fact that in the age of GenAI, students must be evaluated on a broader range of abilities, possibly including on-the-spot viva voce, impromptu presentations, or real-time analysis exercises. Grading criteria must consider parameters beyond just traditional correctness. Evaluations should consider creativity, depth of thought, reading and writing ability, precision, language usage, fluency of speech and the extent of GenAI use.

Employing scenario-based assessments (including AI-generated scenarios) could enable students to critically analyze complex problems in real time. Tools like Turnitin and other AI detection systems could help evaluate degree of originality. Using Bloom's Taxonomy could help draw assessment designs covering multiple cognitive levels, from understanding to critical evaluation. Additional measures could include the application of adjustable testing systems that use AI to modify question difficulty based on student performance, and integrating GenAI tools into the assessment process, for example through AI-generated case studies, AI-enhanced quizzes or peer-group evaluations of AI outputs to develop dynamic and personalized evaluation formats.

Students may also be trained through tasks involving the evaluation of accuracy, bias and reliability of AI-generated content. Such exercises would help develop their critical thinking and ethical judgment.

The assessment strategies discussed above could better prepare learners to face real-world challenges in the age of GenAI.

An academician with 33 years of experience shared that:

It will be difficult to conduct the routine teaching and evaluation process once Gen AI has been introduced. The system has to change accordingly to incorporate the Gen AI. If an assignment is given to a student during the presentation, they should be questioned on justification, interpretation and application. On-the-spot oral questioning like viva voce will help assess whether the student has done the assignment truthfully or just copied it. How a student narrates their work will clarify the integrity of the work. The formative regular assessment might not be suitable. Rather, a more on-the-spot presentation or analysis will help. A punitive grading strategy must be avoided when found guilty of using Gen AI and when no originality is found; rather, helping the students use Gen AI in the way it should be used with their creativity would be fruitful. The grading system will have to be broken accordingly to include the use of Gen AI. [R09]

The plausible grading strategy should be based on creativity, thought process, reading, writing, precision, use of language, flow of speech and amount of Gen AI used. Gen AI cannot be avoided, and we have to include it in the assessment process as to how efficiently Gen AI has been used in an assignment by a student. Whether there was an encroachment in academic integrity, such as honesty, trust, fairness, respect, responsibility, and courage, should be assessed using specialized applications. If encroached, what percentage is acceptable should be delineated. [R01]

We can have scenario analysis- We can provide AI-generated scenarios related to business problems. After that, we can assess students on their ability to analyze, strategize, and propose solutions based on AI-provided data. Further, we can also use AI-generated adaptive assessments that dynamically adjust based on students' responses, providing tailored challenges and content. [R05]

5.4 Dimension 4: Generative artificial intelligence's impact on academic integrity

Theme 1: Long-term policies

Themes 2: Short-term policies

Respondents mentioned that the impact of GenAI on academic integrity policies could be addressed through curricular integration and innovative reforms in policy. Key measures include integrating GenAI tools into the learning process, encouraging collaborative policy design between academia and industry and drawing assessment designs aligning with higher-level thinking skills as put forth by Bloom's Taxonomy. Also, clearly defined academic norms must direct the use of GenAI to generate learning material.

Short-term policy measures could include the introduction of plagiarism and AI-detection tools, training in AI ethics (see, e.g. [Varma et al., 2022](#)), raising awareness about AI-enabled academic misconduct and mandatory disclosure of the extent of GenAI usage in assignments (e.g. a percentage-based declaration) by students. Long-term strategies, on the other hand, could focus on formally integrating AI literacy into the curriculum, encouraging learning and research assisted by AI tools and collaborating with AI industry experts to develop appropriate academic practices. Educational institutions may also lay down the acceptable GenAI use limit, and the authenticity of student work may be questioned if the limit is breached. With the higher objective of preserving genuine, critical and independent student thinking, faculty-members must be intensively trained in assessment tools and applications to monitor and evaluate the use of GenAI. Assessments must focus on higher-order cognitive demands so that GenAI tools do not replace student thinking.

Some quotes are mentioned as follows:

While lower order variables (in Bloom's framework) may get answered by AI, the higher order variables evaluation should be designed to mimic the real-life conceptual 'application-based'

scenario. Incorporation of these policies can enable educational institutions to balance between leveraging the benefits of Generative AI without compromising on academic integrity and ethics. [R7]

The advent of Gen AI and the rapid expansion of its usage calls for the comprehensive revision of academic integrity policies to accommodate its far-reaching implications. The short-term policy measures could involve framing usage guidelines, introducing plagiarism detection mechanisms, training in AI-related ethics, and sensitization to AI-related academic misconduct. The long-term policies could involve the integration of AI literacy into the curriculum, embracing AI-assisted learning, conducting AI-powered research, and collaborating with AI industry experts to stay updated on AI- advancements. Incorporation of these policies can enable educational institutions to balance between leveraging the benefits of Generative AI without compromising on academic integrity and ethics. [R14]

Gen AI will revoke the academic integrity policies in higher education, and it cannot be avoided. Hence, new formulations must be derived with Gen AI's inclusion to maintain academic integrity. For short-term management, we can ask the students to comply with the quantity of Gen AI used in the given assignment without threatening them with punishment. For the long term, an acceptable percentage of usage can be fixed beyond which the authenticity of the work becomes invalid. To formulate the policies with Gen AI inclusion, the authorities of academia should be well-versed in the use of Gen AI. They need intense training and assessment gadgets such as applications to check the overuse of Gen AI. Without which handling Gen AI use by the students would become rampant which would hinder the natural learning process, further making the academic learning process useless. [R17]

Taking students' perspective into cognizance,

Students expressed their belief in responsible and transparent AI use. They mentioned that they recognized the potential of these tools and looked forward to integrating them responsibly into their academic journey. Students stated that they knew these tools did not have the ability to generate original ideas or arguments, and that they viewed them as tools for completing their tasks, such as writing. They highlighted that while GenAI had simplified certain aspects of daily life, its impact on creativity was multilayered. Students felt that while on the one hand, relying solely on AI-generated content could restrict original thinking and self-expression, strategic and intelligent use of AI tools could improve efficiency or help manage repetitive tasks on the other.

Students emphasized the importance of striking a balance between using GenAI to streamline academic tasks and preserving and exercising their own creative capabilities. They believed that such a balanced approach would help them get the benefits of GenAI without compromising intellectual growth or originality.

One of the second-year MBA students mentioned:

Using GenAI writing tools like QuillBot for initial grammar check helped me refine my sentence structure, identify areas for improvement, and eliminate typos. However, I understand that they are unable to generate original ideas or arguments. Instead, I view them as tools to polish my writing, ensuring it reflects my unique voice and perspective. [R48]

The integration of GenAI offers many benefits but raises nuanced ethical considerations. On the positive side, GenAI facilitates personalized learning pathways, unlocking valuable insights from vast amounts of data which leads to increased efficiency, improved student engagement and enhanced educational outcomes. However, the ethical implications of using GenAI must be carefully examined and addressed; for example, overreliance on the usage of GenAI can reduce critical thinking ability and research skills:

Some students use GenAI to write their whole assignments, this will only help them in the short term, but their ability to think will reduce, and they will face challenges later in their professional life. Over-dependence on GenAI tools might impede creativity and critical thinking skills. Though GenAI can provide valuable assistance, students must balance its use with creative exploration

and problem-solving opportunities. Relying too heavily on GenAI for all tasks may limit innovative thinking and personal expression opportunities. [R56]

According to students, problem-based learning, simulation exercises, collaborative projects, gamification, peer assessment, etc. can be the most effective designs for integrating GenAI into the assessment process while preserving creativity.

6. Discussion

The findings with reference to RQ1 revealed that the era of AI-enabled tools in education has arrived, and that it is here to stay. AI is not a passing trend, but a structural transformation that will, if it has not already, most likely permanently change the world around us, our daily lives and everything we do therein. AI will pervade every aspect of our existence, and attempting to ban or suppress AI tool usage in the domain of education would not be realistic. An intelligent approach would be to employ these technologies in educational establishments in ways that stimulate and assist student creativity and not stifle it. Institutions must foster a learning environment where human thought and GenAI tools collaborate meaningfully.

The topic under study and the conflicts associated therewith can be better understood through the lens of Paradox Theory, which highlights the persistent conflicts and oppositions between opposing requirements and points of view. In the context of our study, this conflict between dualities may be seen between innovation and academic integrity, AI adoption and critical human thinking, acceptance of AI systems and control on use of AI tools, etc. Findings show that rather than wholly choosing one side and completely denying the other, educational institutions must manage these conflicts through deliberate, context-sensitive practices. Paradox Theory also emphasizes management rather than one-sided resolution of conflicts. Again, it must be considered that while GenAI offers several unprecedented opportunities to enrich learning experiences, improve collaboration, increase effectiveness of delivery of teaching material and enhance access to knowledge, the same tool can restrict originality, critical thinking and ethical standards if used irresponsibly. Therefore, these conflicting elements must be carefully managed, and the temptation to adhere to personal biases for or against any one point of view must be avoided.

From a Complexity Theory perspective, GenAI integration into education aptly reflects a nonlinear, adaptive process formed by the interaction of several interlinked variables such as institutional policy, preparedness of faculty members, student behavior and rapid advancements in technology. Outcomes cannot be predicted only on the basis of the level of adoption, control or intent; rather, they would be determined by the interaction between humans and technological tools within the environment of a given educational ecosystem. In light of the above, it is clear that a one-size-fits-all approach would neither be prudent, nor effective. The method must be comprehensive and consider views of multiple stakeholders. It must consider the purpose of GenAI in pedagogy, its potential benefits, as well as associated risks, including issues related to bias, exclusivity and data-driven limitations inherent in many GenAI systems that are trained on historical data sets.

Moreover, findings highlight the need to instill ethical values such as fairness, accountability, transparency and privacy into the usage and teaching of AI tools. Responsible use, appropriate attribution and self-discipline must be emphasized. It must be impressed upon students that long-term adherence to values like integrity and honesty will shape their character and also help nurture a more ethical academic and social environment. Notably, institutional leaders, such as directors, deans and faculty members expressed deep concern during the interviews about the lack of clear pedagogical frameworks for integrating GenAI and an absence of reliable assessment tools to help differentiate between human-generated and AI-generated content (Else, 2023; Gonsalves, 2024). Some respondents

went so far as to characterize GenAI tools like ChatGPT as a “threat” to traditional education models. These views highlight the urgent need to redefine learning and evaluation strategies rather than resist the almost inevitable technological change. The need for open and explicit discourse with students about the role of GenAI in their academic journey is a key recommendation that emerged in the study. Respondents laid stress on training students in honest and constructive AI usage that adhered to institutional policies and ethical guidelines (Stokel-Walker and Van Noorden, 2023). Furthermore, disclosure mechanisms like declaring the extent of GenAI usage in assignments could enable greater transparency.

Analysis of responses to *RQ2* indicated that effective student training in GenAI could be achieved through the development of a structured, curriculum-integrated course that emphasizes real-world applications. The course could include foundational concepts such as AI algorithms, machine learning and natural language processing, and stress practical use cases in the domain of business and management. The course would be aimed at imparting skills (ability to understand problem domains, define clear goals, craft well-structured prompts, etc.) essential for operating AI tools to ensure meaningful outputs.

This educational approach aligns with the tenets of Paradox Theory, which underscores the need to manage conflicting requirements. On the one hand, GenAI tools can streamline decision-making and support efficiency; on the other, their use warrants careful judgment, creativity and consideration of ethics. Training students in prompt engineering, for example, represents this paradox where the structure of machine interaction must coexist with human insight. While learning, experimenting and employing AI in their daily tasks, students must learn to manage the dichotomy between automation and originality, efficiency and exploration, self-sufficiency and dependence to attain effective and responsible outcomes.

Complexity Theory in this context guides the design and delivery of such a course. Following the essence of the theory, the course must adopt a nonlinear and adaptive method of learning in AI-rich environments. The level of students’ understanding, comprehension and engagement differs with background, interest and familiarity with technology. Thus, the course must be designed to deliver customized instructional content and exercises addressing the varied learning needs and progression levels of different learners. This method stands in contrast with a one-size-fits-all instruction system, featuring a flexible and student-centered approach and abiding by the principles of complexity and emergence.

Interdisciplinary collaboration in the form of academic-industry partnerships could drive holistic and contextualized learning. Exposing students to different perspectives and real-world applications develops adaptive thinking and enhances their capacity to work across boundaries. However, while fostering a dynamic learning ecosystem, ethical considerations must not be ignored. Regular discussions on fairness, accountability and responsible use throughout the course would go a long way in instilling among students social and ethical values with respect to GenAI use in business and personal domains.

Analysis of responses to *RQ3* provided insights about assessment designs, which should be in sync with teaching content and methodology. Appropriate assessment designs would involve time-bound, individual or team projects with the objective of evaluating students’ ability to solve real-world problems through AI applications. Students can in addition be tasked to evaluate the accuracy, bias and reliability of AI-generated content sharpening their ability for critical analysis. Thought provoking debate sessions can be organized on intellectual honesty, issues pertaining to ethical dilemmas on AI usage. AI-enhanced quizzes and exams can assess students’ comprehension of AI-generated content. Faculty-members may also consider Bloom’s taxonomy for designing assessments, so that the evaluation of higher order variables involves mimicking real-life conceptual scenarios. Contours of the answer may be completed by GenAI, however, the analysis and evaluation

part will require attentive human effort. They should also be aware of possible errors that GenAI in its current form can make (Peres *et al.*, 2023). Teachers can deploy these tools for learning augmentation while harnessing their strengths and weaknesses and enhancing the learning experience.

These findings again align with Paradox Theory and Complexity Theory. In line with Paradox Theory, the recommendations highlight the dichotomy between using GenAI as a helpful tool and preserving academic integrity, critical thinking and ethical judgment. Students are expected to use AI-generated content, but at the same time, question its accuracy, bias and reliability. This paradox aptly depicts both reliance on and scrutiny of AI technology. Complexity Theory is reflected in the call for varied, real-world and higher-order assessments that respond to the unpredictable and developing nature of AI in education. The inclusion of flexible and multidimensional assessment formats like debates, AI-enhanced quizzes and project work indicates the acknowledgement of diverse student needs and the necessity for corresponding customized solutions as opposed to applying a fixed or linear evaluation model.

Response analysis for RQ4 revealed that the advent of GenAI and its widespread usage has resulted in an urgent call for a comprehensive revision of academic integrity policies to accommodate its far-reaching implications. Some short-term policy measures could involve defining AI usage guidelines, employing plagiarism detection mechanisms, training in AI-related ethics and raising awareness about academic misconduct. Long-term policies could involve integration of AI literacy into the curriculum, fixing an acceptable percentage of usage beyond which authenticity of the work becomes invalid, conducting AI-powered research, embracing AI-assisted learning and collaborations and interface with AI industry experts so as to stay abreast of AI related advancements. Such policy formulation would require that the authorities of academia should be well versed in the use of GenAI. Such a policy overhaul would enable educational institutions to achieve a balance between leveraging the benefits of GenAI and maintaining academic integrity.

The students on the other hand must learn the laws and regulations of their universities/institutions. They need to acknowledge that human creativity and intelligence are a pivotal part of writing essays or theses, while AI tools can assist with ideation and a basic text structure. Students may, however, use systems as learning enhancement tools to make their learning process more efficient. Thus, in the long term, a multipronged approach is required to integrate AI technology interventions, education, policy reform and assessment restructuring to address the threats posed by GenAI without compromising academic integrity.

6.1 Implications for teachers and professionals

6.1.1 Focusing on learning objectives. Teachers must contemplate upon the core learning objectives of their courses. In higher education, structured thinking and critical analysis are essential skills. These qualities can be cultivated through deliberate engagement with AI-generated outputs where students critique, revise and refine content using their own thinking and judgment. Such activities would enhance students' logical structuring, critical reflection and problem-solving abilities.

In this context, prompt engineering has been identified as a foundational skill. As the quality of GenAI outputs significantly depends on clear and intelligent prompts, teaching students to craft domain-specific prompts assumes much importance. Similarly, in professional settings, managers must have a clear notion of the specific knowledge outcomes they seek for their teams, and skillfully and intelligently integrate GenAI tools to achieve their objectives.

6.1.2 Generating learning materials using generative artificial intelligence. GenAI tools can help teachers create learning materials that match each student's way of learning and their current level of understanding. For example, teachers can use AI tools (e.g. YouTube

Summary with ChatGPT) to quickly turn videos or lectures into easy-to-understand study notes. This saves time and makes it easier to adjust lessons based on students' needs.

6.1.3 Designing artificial intelligence-supported assessments. Instructors can use GenAI to create quizzes and tests that match what their teaching content. For accuracy and relevance, they should first feed GenAI systems their transcripts or structured lesson plans, and then critically review the generated questions to eliminate errors, bias, or unfair content. Institutions should encourage teachers to integrate GenAI-assisted assessments with human validation processes to ensure rigor and reliability.

Schools should support teachers by combining AI-generated tests with proper human checking, so the assessments stay accurate and trustworthy.

6.1.4 Augmenting learning through generative artificial intelligence. GenAI tools can help teachers and trainers solve difficult teaching problems. They also help students apply their knowledge in real-world situations, check how well they understand concepts and think more critically (Mollick and Mollick, 2023). By using these tools intelligently, with the knowledge of such tools' strengths and limitations, teachers can improve student learning.

There are many ways to include GenAI in lessons. For example, students can review how AI explains a concept in a new situation and suggest ways to improve the response. They can be asked to rewrite an AI-generated essay by changing the prompts or adding new details. Another option is to give students assignments where they improve AI content by adding accurate information from different sources (Mollick and Mollick, 2023).

6.1.5 Promoting responsible and creative use of generative artificial intelligence. Teachers should encourage students to use GenAI as a helpful tool that supports their thinking and writing, not as a crutch to fully depend on. Students should learn responsible and effective use of AI tools; they must strive to improve their own ideas with the help of AI, rather than copying content from it. Students must also be taught to give credit when using such tools and to make sure the work they submit truly reflects their own understanding. Figure 4, explains the implications for teachers and students.

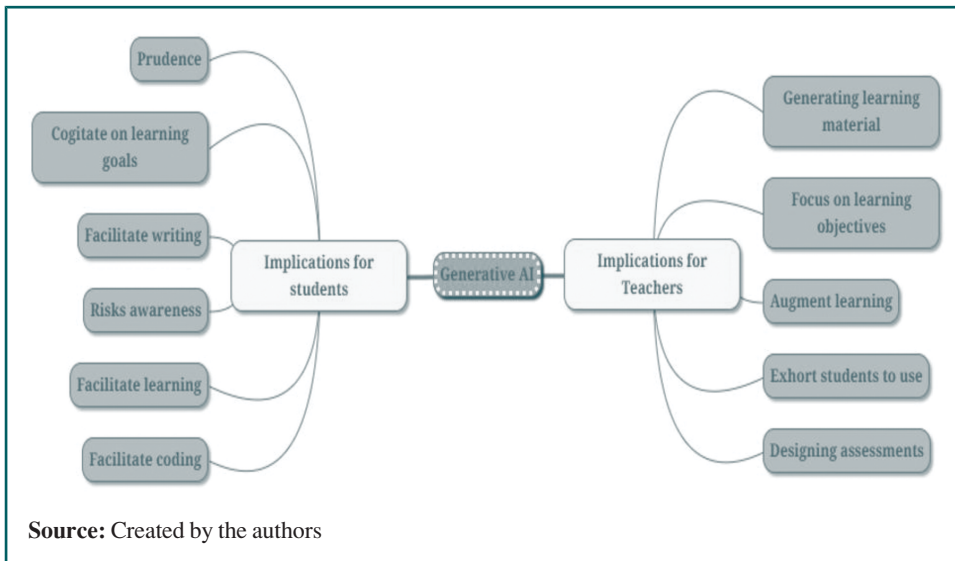
6.2 Implications for students/trainees

Responsive GenAI tools can serve as valuable assets for higher education students, by helping them with academic work, offering accessibility, saving time and refining analytical thinking and linguistic skills (Atlas, 2023). This section reflects on how learners can maximize potential advantages and hedge risks, while using GenAI tools.

6.2.1 Cogitate on Learning Goals. Higher education entails more than just acquisition of domain-specific skills. It includes the development of important skills like independent, critical and structured thinking and clear expression of ideas. Since GenAI tools can create large amounts of well-written text, students must learn to examine that content carefully by questioning its assumptions, analyzing it deeply and improving it through strong reasoning. To get useful answers from GenAI, students must learn how to ask the right questions. This skill, called prompt engineering, is increasingly assuming much importance. Thus, students should think about what they want to learn and how they can use these tools to support and improve their learning process.

6.2.2 Generative artificial intelligence tools as a learning aid. Students can effectively use AI tools to improve their learning under proper guidance. GenAI tools like GPT-4 are already being used in popular educational platforms. For example, the language learning app Duolingo uses GPT-4 to offer features such as Roleplay, where users can practice real-life conversations, and Explain My Answer, which helps them understand why their response was correct or incorrect. These tools make learning more interactive and personalized.

Figure 4 Generative AI and its impact on stakeholders



Another useful feature is the Regenerate option, which lets users request multiple versions of an AI-generated response. When students combine this with carefully crafted, detailed prompts, they can get clearer and more accurate content suited to their learning needs. Students can also use browser extensions that connect with GenAI tools. For example, plugins available on Chrome, Firefox and Safari allow them to extract transcripts from YouTube videos and generate concise summaries. These summaries help in reviewing lectures or tutorials quickly, saving time and helping with revision. Such tools are particularly helpful for students who prefer learning visually or need to revisit material at their own pace. As long as students are taught to use these tools ethically, understand their limitations and learn to verify content independently, GenAI can become a key learning tool in modern education.

6.2.3 Encourage coding with generative artificial intelligence tools. GenAI tools can also aid in generating precise codes. Given the correct prompt, AI tools can provide recommendations for coding (Hauck-Thum *et al.*, 2023). These pieces of code can initiate a project or task. Tools like ChatGPT can assist in debugging code, identifying errors, optimizing them and making them more efficient.

6.2.4 Risks associated with generative artificial intelligence tools. Although GenAI tools can be highly beneficial, their use does carry risks. Both instructors and students must exercise caution against errors and misconduct. ChatGPT, for instance, is known to create non-existent sources, thus propagating false information. Furthermore, there are risks related to plagiarism and copyright infringements, and users must clearly understand their accountability for the originality and correctness of their work.

6.3 Policy and institutional recommendations

Educational establishments can, to a large extent, ensure responsible and creative use of GenAI through carefully designed and effectively communicated regulations and policies. By equipping teachers and students with the tools and frameworks to use GenAI conscientiously and innovatively (see, e.g. Jaiswal *et al.*, 2021), educational institutions can develop and preserve robust academic models, and also foster a culture of ethical innovation, critical thinking and long-term adaptability. Some suggestions are given as follows:

- Integrate GenAI Literacy into core curriculum: Students should complete basic modules covering ethical use, prompt design and critical evaluation of AI output.
- Establish disclosure policies: Institutions must make it mandatory for students to indicate the extent and nature of GenAI use in assignments, similar to citation policies.
- Develop AI-aware assessment frameworks: Blended evaluations must be encouraged; assessment processes must appraise both human and AI-assisted work, using revised rubrics focused on originality, insight and ethical use.
- Train faculty-members in AI-augmented pedagogy: Professional development programs must be held to enable teachers to create GenAI-enhanced content and assessments.
- Set institutional guidelines on acceptable AI use: Clear definitions of what constitutes ethical and unethical use of GenAI in coursework must be established and communicated through official documents and events such as student handbooks, course syllabi and orientation programs.
- Establish AI integration labs or learning units: Innovation hubs/cells could be established within institution premises where educators and students can jointly experiment with GenAI tools under guidance and supervision.

6.4 Limitations and future research

This study adopted a dual-theoretical framework drawing from Complexity Theory and Paradox Theory to better understand the multifaceted implications of integrating GenAI into management education. The study used a qualitative approach and relied on responses from respondents that chose to participate in the study. It considered data collected from different stakeholders across Indian management education institutions. The sample provided valuable insights into the various aspects of GenAI adoption. However, it may not statistically represent the wider academic landscape, and its qualitative and context-specific nature limits statistical generalizability. Furthermore, as educational, cultural, and regulatory environments differ widely across regions, the responses may not represent the views of all educators in the domain. Also, the number of participants for each demographic category does not necessarily represent their respective populations. This study mainly focused on Indian context (i.e. from the top-ranking management institutes) with limited cross-comparison to international studies, therefore the differences in operational, cultural and regulatory contexts may limit the generalizability of the findings of this study. Future studies (e.g. [Anderson et al., 2025](#); [Yusuf et al., 2024](#)) could include larger and more diverse samples from a wider global perspective to establish external validity and comparative value of findings.

Also, studies could focus on evaluating how GenAI tools are integrated and regulated in specific knowledge domains, for example subjects in physical sciences and social sciences. Furthermore, comparative studies across subjects could be undertaken to determine how disciplinary norms shape GenAI usage and regulatory practices. Short and long-term longitudinal studies could also be conducted to identify the progress and changes in GenAI integration and regulation in specific domains in educational institutions.

Finally, this study was conducted in the Indian context only. While India features a unique educational and regulatory environment, findings drawn from the Indian context might not be extendable entirely to other geographical regions with different academic, technological, or regulatory systems. Therefore, caution must be applied while interpreting the findings of the study, particularly in contexts different from that of the study. Future research could be

carried out in other countries and with larger and more diverse sample sizes to generalize results.

Future research may also test the dual-theory framework across regions and disciplines to compare patterns of AI adoption and regulation in diverse academic environments and systems. Longitudinal and interdisciplinary studies may be conducted to offer a deeper understanding of how GenAI shapes learning cultures over time. Moreover, studies could be conducted with particular focus on constructs such as intelligence, ethics and sustainability in relation to GenAI integration into education.

7. Conclusion

The present study sought to explore how management educators experienced and responded to the growing presence of GenAI. Through the lenses of Complexity Theory and Paradox Theory, the study offered a better understanding of the conflict that educational institutions today faced, in terms of acceptance vs resistance of GenAI, highlighting that institutions and educators are confronted by a pedagogical paradox where GenAI can simultaneously undermine academic integrity and offer transformative educational value. This study conceptualizes GenAI integration into management education as an evolving system of interlinked variables and intelligently managed contradictions. Findings demonstrate that GenAI adoption demands balance between innovation and academic integrity, efficiency and creativity and automation and human agency (Eke, 2023).

Existing literature has largely emphasized the capabilities, limitations and ethical concerns surrounding GenAI, albeit in linear terms. The present study fills a major gap in existing literature by emphasizing that the educator's position and experience with GenAI is better understood as a complex and dynamic interplay of conflicting pressures (acceptance vs regulation, efficiency vs learning process, innovation vs authenticity, etc.). Complexity Theory allows us to appreciate the evolving, unpredictable nature of GenAI's role in education (Donaldson *et al.*, 2025; Kolt *et al.*, 2025), while Paradox Theory highlights the importance of embracing contradictory demands without being limited to either/or thinking (Lim *et al.*, 2023; Schaap and Vanlommel, 2024). Through this lens, the apparent dilemmas are not problems to be solved, but tensions to be managed and balanced over time. With the proliferation of GenAI tools and their increasing integration in educational practices, the dimensions under consideration go beyond simply technological and come to encompass ethical, pedagogical and philosophical aspects as well. By adopting a balanced approach toward AI integration and regulation in education and formulating context-sensitive strategies, academia can shape the GenAI paradox into a more productive educational innovation (see Appendix).

Critically, the study offers significant originality value to the state of the art by (i) contextualizing GenAI adoption in a non-Western setting, thereby expanding the scope of existing literature; (ii) employing a dual-theory lens (that has hitherto remained unused in this context) to explain institutional responses to disruptive technologies and (iii) converting theoretical insights into actionable policy, and pedagogical and governance frameworks for responsible AI integration. These contributions offer a structured analytical model for global educators and policymakers to adopt and follow.

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Data availability

Data can be made available upon reasonable request.

Credit author statement

Shalini Nath Tripathi: conceptualization, validation, writing – original draft, editing and reviewing; Nishtha Malik: conceptualization, writing – revised draft, investigation, methodology, data curation, formal analysis, visualization, writing – original draft; Himanshu Rai: conceptualization, investigation, supervision, visualization, validation, writing – editing and reviewing; Arup Varma: conceptualization, validation, writing – editing and reviewing.

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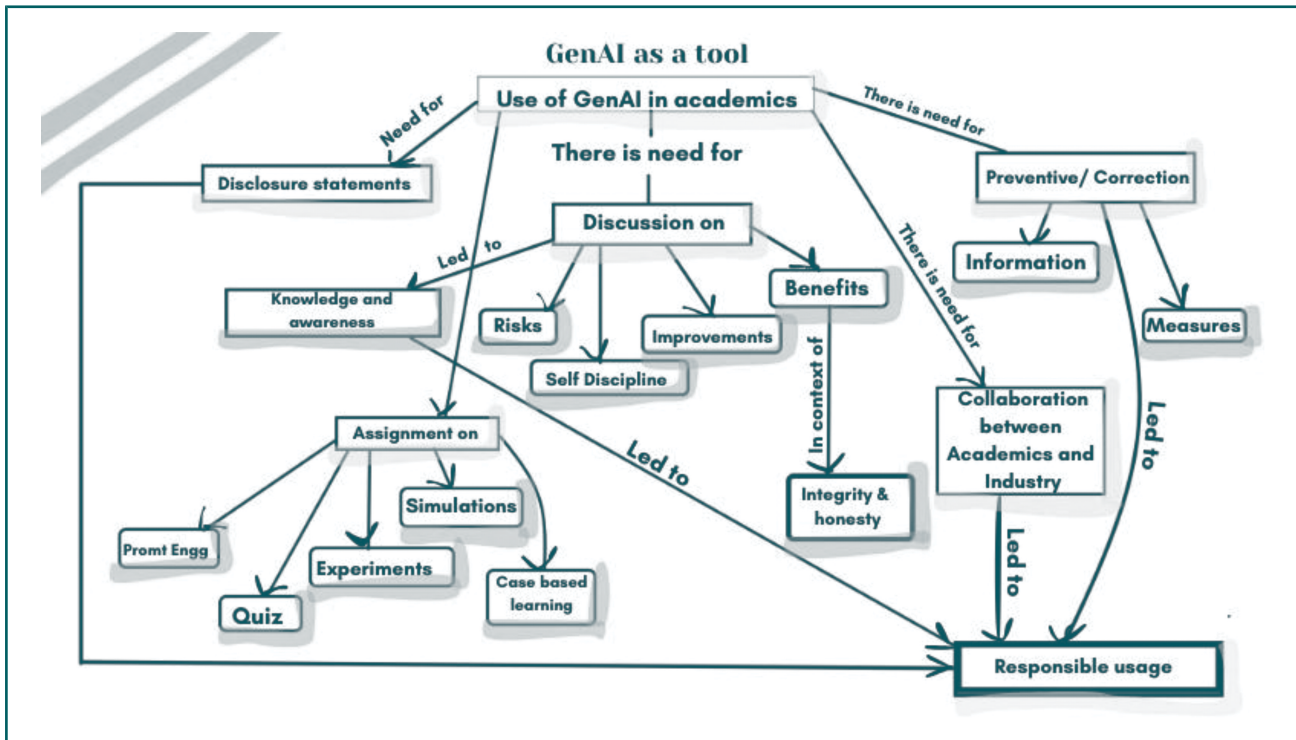
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Further reading

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Figure A1 Concept Map

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